

CORA: A Protocol for Diagnosing Boundary Robustness in Text-to-Audio Retrieval under Query Reformulations

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Abstract

Text-to-Audio (T2A) retrievers are typically evaluated with caption style queries, but the same user intent can be expressed in many forms. We introduce CORA (Caption-Offset Retrieval for Audio), a caption anchored diagnostic protocol that rewrites each source caption into five intent preserving forms (Command, Question, Indirect, Key phrase, and Statement) while fixing the target audio. By tracking the same target across query forms, CORA defines RankDrop, a metric revealing failures hidden by Recall@k. Using Pearson’s correlation coefficient r , we find that RankDrop is weakly associated with raw text space movement ($r = 0.084$), but strongly associated with Target Alignment Loss and Target Boundary Margin Degradation ($r = 0.508$ and $r = 0.615$). The same pattern appears in OEA retrievers, where RankDrop is better explained by boundary degradation ($r = 0.472/0.478$) than by query movement ($r = 0.084/0.046$). Overall, these results suggest that robust T2A retrieval requires preserving the target’s boundary advantage over competing audio under reformulation.¹

1 Introduction

Text-to-audio (T2A) retrieval systems are typically evaluated with caption style text queries. However, the same semantic intent can be expressed in substantially different linguistic forms including imperative command, a yes/no question, a polite indirect request, a keyword phrase, or a declarative statement. A robust retriever should preserve the target audio across these intent-preserving reformulations, but standard Recall@k evaluations provide only aggregate performance and cannot reveal whether the same target experiences a rank decline when the query form changes.

Query variation can substantially affect retrieval robustness in text information retrieval (Penha et al., 2022), and real sound search queries differ markedly from the caption style descriptions commonly used in T2A benchmarks (Weck and Font, 2024). RobustCLAP addresses linguistic variation through multi-view contrastive learning (Selvakumar et al., 2025), while OEA introduces user intent queries and Hard Negative oriented evaluation for audio-text retrieval (Yoo et al., 2026). These studies show that query form robustness matters and that dedicated training or evaluation can improve it. In this work, we ask a complementary diagnostic question: when retrieval failures persist despite intent preserving reformulation, what embedding space change drives them?

We introduce CORA, a diagnostic protocol anchored to source captions for T2A retrieval under expressive query reformulations. Given a benchmark caption, CORA generates five semantically aligned query forms (Command, Question, Indirect, Key phrase, and Statement) while keeping the target audio and candidate pool fixed. CORA tracks the rank of the same target audio before and after reformulation, revealing that reformulation effects depend on the model–dataset regime: a query form that is stable in one setting can be harmful in another.

We then analyze the mechanisms behind these rank drops. We compare a text space movement account with a target boundary account, which examines the target’s score advantage over competing non target audio. Across 66,100 CORA pairs, raw query movement is only weakly associated with RankDrop ($r = 0.084$). Conversely, Target Alignment Loss (TAL) and Target Boundary Margin Degradation (TBMD) show much stronger associations ($r = 0.508$ and $r = 0.615$). Notably, this boundary degradation pattern persists in external robustness oriented retrievers like OEA and RobustCLAP, indicating that preserving the target margin

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¹We release our code and the CORA dataset at <https://github.com/sogang-isds/CORA>.

is a core aspect of stable retrieval.

Our contributions are as follows:

- We introduce CORA, a caption anchored diagnostic protocol for T2A retrieval that fixes the target audio and candidate pool across five intent preserving reformulations.
- We show that query form sensitivity is not determined by surface form alone, but depends heavily on the model and dataset regime.
- We provide paired rank diagnostics demonstrating that residual retrieval degradation (even in robustness oriented retrievers) is better characterized by target alignment loss and boundary degradation than by raw text space movement.

2 Related Work

2.1 Audio-Language Models

Contrastive Language-Audio Pretraining (CLAP) has become the dominant paradigm for T2A retrieval, learning joint embedding spaces via InfoNCE loss (Wu et al., 2023; Elizalde et al., 2023). Scaling along data and model axes has yielded consistent gains (Mei et al., 2024; Microsoft, 2023), while a complementary line of work addresses structural failures within the standard alignment space: Ghosh et al. (2024) exposed limitations in compositional reasoning; Li et al. (2024) introduced locality-aware features to recover fine-grained correspondences; Niizumi et al. (2025) combined masked modeling with contrastive pre-training; and Yuan et al. (2024) proposed temporal-contrastive captions to recover ordering information. These contributions illuminate representational limits of CLAP, but leave open why retrieval degrades under linguistic variations of the same semantic intent — the question CORA is designed to answer.

2.2 Query Robustness in Audio and Text Retrieval

Query brittleness is well-documented in information retrieval. For instance, Penha et al. (2022) showed that neural pipelines suffer roughly a 20% drop in effectiveness under meaning-preserving paraphrases. More recently, Weck and Font (2024) identified a stark distributional gap between real Freesound user queries (averaging just 1.8 tokens) and the lengthy, multi-word captions conventionally used for training and evaluation (Kim et al.,

2019; Drossos et al., 2020). While Selvakumar et al. (2025) mitigated paraphrase-driven performance drops via multi-view contrastive fine-tuning, and Yoo et al. (2026) exposed hidden retrieval limitations using a five-type User-Intent Query (UIQ) taxonomy, these approaches largely focus on aggregate robustification or macro-level evaluation. In contrast, our work adopts a granular, instance-level perspective, tracking whether a specific target remains retrievable under systematic query form shifts. Rather than proposing new training objectives or architectural modifications, we focus on a diagnostic objective: characterizing residual failures through directional geometry in the joint embedding space—a subtle failure signal that aggregate metrics inherently fail to capture.

2.3 Contrastive Embedding Geometry and Hard Negatives

Hard negative mining is foundational to contrastive learning (Robinson et al., 2021), and has been applied to improve compositional reasoning, fine-grained alignment, and text retrieval at scale. Jiang et al. (2024) show that under hard negative sampling, optimal representations converge to neural-collapse configurations, making embedding geometry a precise object of study. In vision-language models, Liang et al. (2022) identified a modality gap in CLIP arising from initialization and contrastive loss dynamics; Kang et al. (2025) proved formally that no CLIP-like space can simultaneously achieve correct attribute binding and compositional understanding. We deploy hard negative as *inference-time diagnostic probes*, measuring whether a query-shift vector moves toward the target and away from the decision boundary between the target and its hard negatives. This signal captures precisely the mechanism by which form shifts displace a target behind its nearest competitor.

3 Method

3.1 Caption-Offset Retrieval for Audio (CORA)

Most existing audio-text retrieval benchmarks rely on caption-style queries, yet real users frequently express the same intent as commands, questions, indirect requests, keyword phrases or declarative statements. To evaluate robustness to diverse user formulations, we define five query types commonly observed in real-world audio retrieval scenarios: Command, Question, Indirect, Key phrase, and

Statement queries.

Our taxonomy is based on the query formulations introduced in UIQ (Yoo et al., 2026), but we modify several query types and generation strategies to better reflect the diversity of real-world user expressions (Section 3.2). CORA uses this taxonomy to evaluate retrieval robustness. Sections 3.3–3.5 define the paired evaluation and the embedding-space diagnostics that make up the actual protocol. Full query definitions and examples are provided in the Appendix A.

3.2 CORA Generation

Unlike UIQ, CORA explicitly introduces *Indirect* queries as an independent interaction type. In realistic retrieval settings, users often employ indirect formulations such as “*Could you help me find ...*” or “*I’m trying to locate ...*”.

In addition, UIQ relies on fixed opening patterns for each query type, such as “*Can you ...*” for Question or “*Locate ...*” for Command. Such patterns constrain the query distribution to narrow lexical forms, making it difficult to evaluate robustness to realistic formulation variation. CORA is designed to encourage broader formulation diversity across query types. We additionally verify that generated queries preserve the semantics of the original source captions. Detailed generation procedures, prompts, diversity analyses, and verification results are provided in the Appendix A.

3.3 Metric

We evaluate CORA reformulations by tracking how the rank of the same target audio changes when the source query is replaced by a CORA query. For each CORA pair $p \in \mathcal{P}$, let $\text{rank}_{\text{src}}(p)$ and $\text{rank}_{\text{cora}}(p)$ denote the 1-indexed rank of the target audio under the source query and the CORA query, respectively. Both ranks are computed by encoding the corresponding query directly with the retriever’s text encoder, with no additional preprocessing or query rewriting applied beforehand. We define rank drop as

$$\text{RD}(p) = \text{rank}_{\text{cora}}(p) - \text{rank}_{\text{src}}(p). \quad (1)$$

Positive values indicate that the target is ranked lower after reformulation, while negative values indicate that its rank improves. We report the mean over all evaluated CORA pairs:

$$\text{MeanRD} = \frac{1}{|\mathcal{P}|} \sum_{p \in \mathcal{P}} \text{RD}(p). \quad (2)$$

While Recall@5 uses a hard top- K threshold that masks severe absolute rank fluctuations (e.g., dropping from rank 6 to 100), MeanRD performs a paired, instance-level analysis.

3.4 Embedding-Space Diagnostic Metrics

Let q_{src} and q_{cora} denote the text embeddings of the source query and its CORA reformulation, respectively, and let a_{tgt} denote the ground-truth target audio embedding. We use cosine similarity as the retrieval score.

Query movement. We first define a text-space movement baseline:

$$\Delta_{\text{move}} = 1 - \cos(q_{\text{cora}}, q_{\text{src}}). \quad (3)$$

This metric measures how far the CORA query moves from the source query in the text-embedding space.

Target margin. To measure target-margin degradation, we compare the target audio with the hard negative for each query, defined as the highest-scoring non-target audio:

$$\begin{aligned} h_{\text{src}} &= \arg \max_{a_j \neq a_{\text{tgt}}} \cos(q_{\text{src}}, a_j), \\ h_{\text{cora}} &= \arg \max_{a_j \neq a_{\text{tgt}}} \cos(q_{\text{cora}}, a_j). \end{aligned} \quad (4)$$

We define the target margin for query q and hard negative h as

$$M(q, h) = \cos(q, a_{\text{tgt}}) - \cos(q, h). \quad (5)$$

A larger margin means that the target receives a higher score than its hard negative by a larger amount. We then define Target Boundary Margin Degradation (TBMD) as the reduction in this margin after reformulation:

$$\text{TBMD} = M(q_{\text{src}}, h_{\text{src}}) - M(q_{\text{cora}}, h_{\text{cora}}). \quad (6)$$

A larger TBMD indicates that the CORA query reduces the target’s score advantage over its hard negative.

This margin degradation can be decomposed into two terms:

$$\begin{aligned} \text{TBMD} &= \underbrace{\cos(q_{\text{src}}, a_{\text{tgt}}) - \cos(q_{\text{cora}}, a_{\text{tgt}})}_{\text{TAL}} \\ &\quad + \underbrace{\cos(q_{\text{cora}}, h_{\text{cora}}) - \cos(q_{\text{src}}, h_{\text{src}})}_{\text{HNP}}. \end{aligned} \quad (7)$$

The first term, Target Alignment Loss (TAL), measures how much similarity to the target audio decreases after reformulation. The second term, Hard Negative Pull (HNP), measures how much the hard negative becomes more competitive. Thus, TBMD captures both sides of retrieval degradation: reduced alignment with the target and increased competition from a hard negative. We refer to this formulation as dynamic-HN TBMD. Appendix F verifies that it gives a stronger association with RD than two fixed hard-negative alternatives.

3.5 Directional Projection Diagnostics

We additionally analyze the direction of the source to CORA query shift in the joint embedding space. We compare its magnitude $\|d\|_2$ with projections onto the target-audio direction (p_{audio}) and the source anchored target boundary direction (p_{boundary}). These are diagnostic associations with RD, full feature definitions and hard negative selection are given in Appendices K.1 and K.2.

4 Experiment

4.1 Experimental Setup

We evaluate retrieval robustness under the CORA protocol by comparing each source query with five semantically aligned reformulations: Command, Question, Indirect, Key phrase, and Statement. We evaluate four datasets: AudioCaps, Clotho, MACS, and MECAT. For the main diagnostic analysis, retrieval is performed within each dataset candidate pool. Figure 1 and Tables 2 and 3 use a single candidate pool containing audio from all four datasets. These results are reported as a sensitivity analysis.

The main experiments use four CLAP models without additional fine-tuning: LAION-CLAP, M2D-CLAP, MGA-CLAP, and MS-CLAP. For these experiments, we use only the test splits (3,367 items). Excluding 62 items with an incomplete set of query forms leaves 3,305 items and yields 66,100 paired evaluations across five forms and four CLAP retrievers. Table 1 provides the dataset-level counts and filtering details. We additionally apply the same paired protocol to OEA and RobustCLAP in Section 4.5. Full dataset and model details are provided in Appendix B.

4.2 Query-Form Sensitivity Depends on Model-Dataset Regime

Overall form axis. Figure 1 reports MeanRD by query form, averaged across datasets and

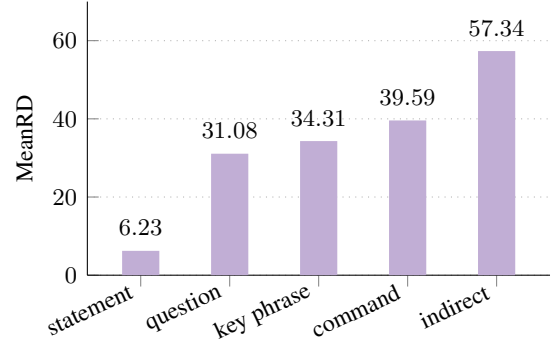


Figure 1: Overall MeanRD by query form, averaged across all datasets and models. Statement queries (most similar to original captions) are the most stable, while indirect queries cause the largest rank drop.

CLAP models. Statement queries show the smallest MeanRD on average, likely because their structure is closest to standard caption-style descriptions. In contrast, indirect queries produce the largest rank drop on average, suggesting that this query form is more difficult under the evaluated retrieval settings.

Dataset-form axis. Table 2 shows that the global hierarchy does not hold uniformly across datasets. For instance, key phrase queries are highly harmful on Clotho, with a MeanRD of 114.92, but actually improve the target rank on MECAT, yielding a MeanRD of -39.55 . Similarly, statement queries are stable on AudioCaps and Clotho, but cause severe rank drops on MACS. These reversals clearly indicate that CORA sensitivity is not determined by surface form alone; a query form that is safe in one dataset can be harmful in another.

Dataset-model axis. Table 3 next decomposes MeanRD by dataset and CLAP model. This analysis shows that CORA sensitivity also depends on the retriever. On AudioCaps, LAION, M2D, and MGA all show positive MeanRD, indicating a rank drop, whereas MS-CLAP shows a large negative MeanRD (-109.96). On MACS, LAION and MS-CLAP experience substantial rank drops, while M2D and MGA show negative MeanRD. MECAT exhibits a different pattern: all four models exhibit rank drops, with LAION showing the largest MeanRD. A complete retriever-level summary comparing aggregate recall and paired RD is provided in Appendix C.

Summary. Overall, these results show that CORA-induced rank drops cannot be reduced to a single difficult query form or a single weak model. The direction and severity of rank change depend heav-

Dataset	Source captions	Validation	Test	Excluded	Retained	Evaluations
AudioCaps	1,469	495	974	11	963	19,260
Clotho	2,090	1,045	1,045	51	994	19,880
MACS	1,000	500	500	0	500	10,000
MECAT	1,008	160	848	0	848	16,960
Total	5,567	2,200	3,367	62	3,305	66,100

Table 1: Dataset counts and filtering for the main CORA evaluation. Source captions include the validation and test splits, whereas retrieval uses only test items. The excluded items have an empty indirect query (11 AudioCaps items) or an empty key-phrase query (51 Clotho items). Each retained item contributes five query forms evaluated with four CLAP retrievers.

Dataset	Query form	MeanRD
AudioCaps	Command	-8.13
	Indirect	11.63
	Key phrase	23.58
	Question	-13.83
	Statement	-21.29
Clotho	Command	-1.07
	Indirect	24.19
	Key phrase	114.92
	Question	7.01
	Statement	-6.49
MACS	Command	14.66
	Indirect	89.99
	Key phrase	20.02
	Question	46.75
	Statement	77.03
MECAT	Command	156.14
	Indirect	128.88
	Key phrase	-39.55
	Question	101.05
	Statement	10.63

Table 2: Dataset-by-form MeanRD. A query form that causes a large rank drop in one dataset can be neutral or beneficial in another, showing strong dependency on the dataset regime.

ily on the interaction between the dataset and the retriever. The complete dataset–model–query–form retrieval breakdown is provided in Appendix D. We further support this interactional pattern with grouped cross-validated prediction in Appendix E.

On the 2,805 items shared with UIQ, we also compare CORA, CORA without diversification, and released UIQ using identical source caption anchors, target audios, dataset candidate pools, and four CLAP retrievers. Their MeanRD values are +5.28, +4.17, and −5.78, respectively. This is a descriptive comparison of query artifacts with different generation procedures and form inventories (Appendix A.5).

Dataset	Model	MeanRD
AudioCaps	LAION	49.63
	M2D	22.79
	MGA	31.10
	MS-CLAP	-109.96
Clotho	LAION	31.92
	M2D	17.00
	MGA	16.37
	MS-CLAP	45.57
MACS	LAION	214.79
	M2D	-80.54
	MGA	-8.15
	MS-CLAP	72.67
MECAT	LAION	187.70
	M2D	11.26
	MGA	68.55
	MS-CLAP	18.21

Table 3: Dataset-by-model MeanRD under CORA reformulations. The direction and severity of rank drops vary across CLAP models within the same dataset, indicating that CORA sensitivity depends on the retriever as well as the dataset.

4.3 TBMD Characterizes Rank Degradation Better than Query Movement

Section 4.2 showed that the same query form can be harmful in one model-dataset regime and neutral or beneficial in another. We therefore ask a more specific diagnostic question: when the target audio drops in rank after CORA reformulation, what kind of embedding-space change explains that degradation? We compare two accounts. The *query-movement account* predicts that RD should mainly increase when the reformulated query moves far from the source query in text space. The *target-boundary account* predicts that RD should increase when the reformulated query weakens the target audio’s score advantage over its strongest non-target competitor. Based on these accounts, we design three diagnostic signals—query movement, Target Alignment Loss (TAL), and Target Boundary Mar-

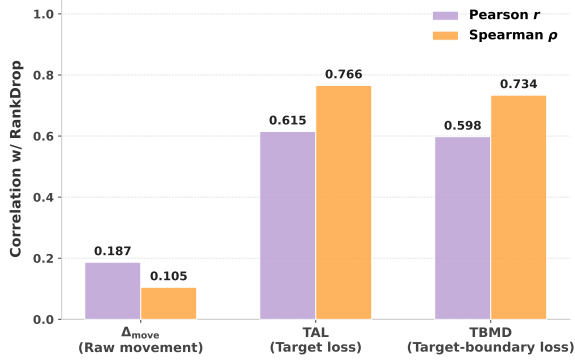


Figure 2: Correlation between diagnostic signals and RD over 66,100 CORA pairs using the pooled cross-dataset candidate set. Target-specific and boundary-specific signals, TAL and TBMD, show substantially stronger association with RD than query movement Δ_{move} .

gin Degradation (TBMD)—and test which signal better characterizes continuous target-rank degradation.

4.3.1 Main Diagnostic Analysis

We evaluate the above hypothesis over 66,100 CORA pairs from four datasets and four CLAP models. For each pair, we run retrieval twice over the same candidate pool: once with the source query and once with the CORA query. This provides the source rank and the CORA rank of the same target audio, allowing us to compute the RD. We then compute the three diagnostic signals: Δ_{move} , TAL, and TBMD from the corresponding text and audio embeddings.

Across the 66,100 CORA pairs evaluated with per-dataset candidate pools, the overall MeanRD is 3.881. As a sensitivity analysis, we also evaluate the same pairs with a pooled cross-dataset candidate set. In this setting, 32,992 pairs (49.91%) show rank worsening and MeanRD is 33.711. The pooled-setting correlations are summarized in Figure 2.

Overall diagnostic result. In the primary per-dataset analysis, raw query movement (Δ_{move}) has only a weak association with RD ($r = 0.084$). This is a crucial finding: simply reformulating a query so that it moves far away from the original caption in text space does not automatically mean the target audio will be lost.

By contrast, the target-boundary signals show much stronger associations with rank degradation. Target Alignment Loss (TAL) and Target Boundary Margin Degradation (TBMD) achieve Pearson

correlations of 0.508 and 0.615, respectively. This supports the target-boundary account: rank degradation is associated with a loss of target alignment or a narrowing of the target’s margin against competitors, rather than with how far the text moves on the surface. Because TBMD is computed from the same similarity scores that determine rank, this association is diagnostic rather than causal; Appendix H reports a control analysis showing that TBMD retains explanatory power over TAL after accounting for dataset, model, query form, Δ_{move} , source rank, and candidate pool size. Appendix G further checks anchor and content controls, and Appendix I supports this interpretation with a token-level audit.

On the matched three-dataset subset, the released UIQ queries show the same ordering: Pearson’s r with RankDrop is -0.075 for Δ_{move} , 0.508 for TAL, and 0.538 for dynamic-HN TBMD (Appendix A.5). This is a descriptive comparison because UIQ differs in generation procedure, grounding, and query-type inventory.

Dataset-level analysis. In the pooled cross-dataset sensitivity analysis, we further examine whether this diagnostic pattern holds across datasets. Here, Pearson’s r measures the linear association between each diagnostic signal and RD: values near zero indicate little association, whereas larger positive values indicate that the signal increases with RD. Across AudioCaps, Clotho, MACS, and MECAT, the association for Δ_{move} tends to be lower or more variable, whereas TAL and TBMD generally exhibit higher positive associations with RD. For example, Δ_{move} shows little correlation with RD on AudioCaps ($r = -0.016$), while TAL and TBMD reach $r = 0.531$ and $r = 0.547$. On MACS, Δ_{move} reaches $r = 0.361$, whereas TAL and TBMD reach $r = 0.737$ and $r = 0.716$. These dataset-level results indicate that the diagnostic pattern is not isolated to a single dataset: RD tends to align more closely with TAL and TBMD than with Δ_{move} . The full dataset-level breakdown is provided in Appendix J.

4.3.2 TBMD-Guided View Intervention

The previous analysis shows that TBMD is consistently associated with RD. We further conduct a controlled view-selection check: for each source caption, we select one of the five CORA reformulations while keeping the target audio, retrieval model, and candidate pool fixed. If TBMD cap-

Selection policy	MeanRD
Average over views	3.881
Min- Δ_{move} view	1.875
Min-TAL oracle	-18.568
Min-TBMD oracle	-19.467
Best-rank oracle	-24.560

Table 4: Controlled CORA-view intervention. Lower MeanRD indicates better target rank preservation; negative values indicate that the selected CORA view improves the target rank relative to the source query. Values in this table are used to compare selection policies under the same controlled setting.

tures the degradation pattern, **Min-TBMD oracle** should yield lower MeanRD than the text-space-closest policy.

Selection policies.

- **All CORA forms**: average over all five reformulations.
- **Min- Δ_{move} view**: choose the text-space-closest reformulation.
- **Min-TAL oracle**: choose the reformulation with the smallest TAL.
- **Min-TBMD oracle**: choose the reformulation with the smallest TBMD.
- **Best-rank oracle**: choose the reformulation with the best target rank.

The **Min-TAL oracle**, **Min-TBMD oracle**, and **Best-rank oracle** are target-aware diagnostic oracles and are not intended as inference-time methods.

Table 4 shows that selecting the **Min- Δ_{move} view** only slightly reduces MeanRD, from 3.881 to 1.875, compared with averaging over CORA views. In contrast, **Min-TBMD oracle** reduces MeanRD to -19.467 , slightly below **Min-TAL oracle** (-18.568) and closer to the **Best-rank oracle** upper bound (-24.560). This is consistent with the diagnostic role of TBMD: compared with TAL alone, TBMD also accounts for whether a hard negative becomes more competitive after reformulation.

4.4 Projection Analysis Validates the Directional Boundary Hypothesis

The previous section analyzed target-alignment change and local-margin degradation using scalar metrics such as TAL and TBMD. However, these scalar quantities do not fully reveal whether the retrieval degradation is caused by the direction of the embedding shift. Therefore, we further analyze whether the embedding shift from the source

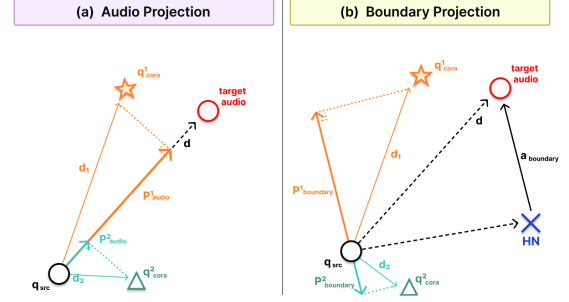


Figure 3: Illustration of (a) the audio-projection and (b) the boundary-projection perspectives for query shift.

caption to the CORA query moves in a direction that preserves or degrades the retrieval boundary between the target and hard negative.

We analyze the relationship between RD and three features: $\|d\|_2$, p_{audio} , and p_{boundary} . Formal feature definitions and hard negative selection details are provided in the Appendix K.1 and Appendix K.2

- $\|d\|$: the magnitude of the embedding shift, indicating how far the reformulated query moves in the embedding space.
- p_{audio} : the projection of d onto the target-audio direction, measuring alignment between the query movement and the target audio embedding.
- p_{boundary} : the projection of d onto the local target boundary direction, measuring whether the query movement preserves retrieval boundary against nearby hard negatives.

Figure 3 illustrates the core intuition behind the projection analysis. Figure 3(a) presents the audio-projection perspective. Although q_{cora}^1 moves farther away from the source caption embedding than q_{cora}^2 , its embedding shift direction d_1 remains better aligned with the target-audio direction, resulting in less retrieval collapse.

Figure 3(b) presents the boundary-projection perspective. Even with larger displacement, q_{cora}^1 remains better aligned with the target boundary direction, thereby preserving local retrieval separation against nearby hard negatives. Meanwhile, q_{cora}^2 weakens the local retrieval boundary despite its smaller $\|d\|_2$.

To evaluate the explanatory power of each feature set, we perform grouped cross-validated Ridge regression for RD prediction. Detailed regression

Feature set	Pearson r	Spearman ρ
fixed effects only	0.174	0.178
+ $\ d\ $	0.206	0.196
+ p_{audio}	0.500	0.594
+ p_{boundary}	0.422	0.529

Table 5: Correlation between projection-based feature sets and RD prediction.

settings are provided in the Appendix K.3 We then compute Pearson and Spearman correlations between predicted and ground-truth RD values.

As shown in Table 5, all regression settings additionally include fixed effects for dataset, retrieval model, and query form. Adding the $\|d\|$ achieves only a weak association with RD ($\rho = 0.196$). In contrast, the projection-based features show substantially stronger correlations: p_{audio} reaches $\rho = 0.594$, while p_{boundary} reaches $\rho = 0.529$.

These results support the TBMD interpretation from a directional perspective: retrieval degradation occurs not when the query moves far from the source, but when the embedding shift fails to preserve target alignment and local boundary separation from hard negatives.

4.5 TBMD Characterizes Residual Rank Degradation in Query-Robust Retrievers

We further test whether the target-boundary pattern observed above also holds for retrievers designed or evaluated for query-form robustness. We evaluate three external retrievers: OEA-Qwen3B-Cl, OEA-Qwen3B-AC, and RobustCLAP. OEA represents user-intent-oriented audio-text retrieval, and RobustCLAP represents robustness-oriented retrieval trained with paraphrastic views of the same audio scene (Yoo et al., 2026; Selvakumar et al., 2025). We ask whether the remaining CORA rank degradation in these models is still better characterized by TBMD than by query movement. To investigate this, we apply the CORA paired evaluation protocol to each external retriever, computing RD and diagnostic correlations independently.

Table 6 shows two patterns. First, OEA variants substantially reduce CORA degradation, with MeanRD close to zero. RobustCLAP shows larger MeanRD in this evaluation, despite being designed for linguistic-variation robustness. Second, the residual rank degradation in all three retrievers follows the same diagnostic ordering as in the CLAP models. Query movement is only weakly associated with RD, with correlations from 0.046

to 0.174. By contrast, TBMD is more strongly associated with RD across all three retrievers, with correlations from 0.472 to 0.508.

This result indicates that models designed for intent-oriented or linguistic-variation-robust retrieval can reduce the severity of CORA degradation, but the remaining errors are still explained by the same target-boundary mechanism. In other words, even after robustness-oriented training or evaluation, residual CORA RD is better characterized by degradation of the target’s margin against its strongest non-target competitor than by the amount of query movement alone.

5 Discussion

CORA should be interpreted as a paired diagnostic protocol. Recall@ k measures aggregate retrieval quality, while RD checks whether the same target audio is preserved after a CORA reformulation. This paired view is important because our results show that query form sensitivity changes across the model and dataset regime, rather than following a fixed ordering of surface forms.

The embedding diagnostics suggest that residual RD is not well explained by raw text space movement alone. TAL and TBMD provide more informative signals because they measure whether the reformulated query weakens target alignment and the target audio’s score advantage over its Hard Negative. These results do not show that TAL or TBMD should be used as inference time objectives. Rather, they indicate which retrieval changes are associated with CORA failures under fixed targets and candidate pools.

This diagnostic perspective clarifies the limits of simple inference time corrections. As shown in Appendix L, interventions like MultiCLAP score fusion can improve aggregate R@5 as a supporting baseline, but they do not resolve the underlying query form brittleness. Appendix M finds a similar result for query preprocessing. Compared with no preprocessing, general-purpose or CORA-specific rewriting lowers MeanRD, but the TAL/TBMD over movement ordering holds under all three conditions. True robustness requires explicitly addressing intent variation during the training stage, as demonstrated by OEA’s reduced RD. Therefore, future robustness methods must be evaluated not just on aggregate performance, but on their ability to preserve a target’s margin over competitive non target audio under expressive reformulation.

Model	CORA R@5	MeanRD	$r(\Delta_{\text{move}}, \text{RD})$	$r(\text{TBMD}, \text{RD})$
OEA-Qwen3B-Cl	43.34	0.603	0.084	0.472
OEA-Qwen3B-AC	41.17	0.479	0.046	0.478
RobustCLAP	35.47	9.408	0.174	0.508

Table 6: External retriever diagnostics under the CORA protocol. CORA R@5 is reported as a percentage. MeanRD measures average target-rank degradation after reformulation. Correlations are computed separately for each retriever.

6 Conclusion

We introduced CORA, a caption-anchored diagnostic protocol for evaluating whether text-to-audio retrievers preserve the same target audio under intent-preserving query reformulations. Across four datasets and multiple retrievers, CORA shows that query-form sensitivity is conditional on the model-dataset regime rather than being an intrinsic property of a particular surface form.

Our main diagnostic finding is that residual RD is more closely associated with Target Alignment Loss and Target Boundary Margin Degradation than with raw text-space movement. The same pattern also appears in external robustness-oriented retrievers, including OEA and RobustCLAP. These results suggest that robust T2A retrieval requires more than invariance to query wording: it should preserve the target audio’s score advantage over competitive non-target audio under reformulation. CORA provides a compact testbed for analyzing this target-boundary behavior and for developing future boundary-aware robustness methods.

Limitations

This work has three main limitations. First, CORA uses LLM-generated reformulations rather than real user search logs; although we verify semantic alignment and query type validity, generated queries may still contain artifacts and underrepresent multilingual queries. Second, CORA is a controlled caption reformulation diagnostic, so it does not cover unconstrained intent, multi-target relevance, or specialized domains such as speech, music, medical, or industrial audio. Third, our findings are based on CLAP models and selected external retrievers, and TAL/TBMD remain diagnostic rather than causal signals whose behavior may change with cross-encoder, autoregressive, or larger-corpus retrieval settings and with different Hard Negative pools.

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References

- Konstantinos Drossos, Samuel Lipping, and Tuomas Virtanen. 2020. Clotho: An audio captioning dataset. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 736–740. IEEE.
- Benjamin Elizalde, Soham Deshmukh, Mahmoud Al Ismail, and Huaming Wang. 2023. CLAP: Learning audio concepts from natural language supervision. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5. IEEE.
- Sreyan Ghosh, Ashish Seth, Sonal Kumar, Utkarsh Tyagi, Chandra Kiran Reddy Evuru, S Rameshwaran, S Sakshi, Oriol Nieto, Ramani Duraiswami, and Dinesh Manocha. 2024. CompA: Addressing the gap in compositional reasoning in audio-language models. In *The Twelfth International Conference on Learning Representations (ICLR)*.
- Steve Han, Gilberto Titericz Junior, Tom Balough, and Wenfei Zhou. 2025. Judge’s verdict: A comprehensive analysis of LLM judge capability through human agreement. *arXiv preprint arXiv:2510.09738*.
- Ruijie Jiang, Thuan Nguyen, Shuchin Aeron, and Prakash Ishwar. 2024. Hard-negative sampling for contrastive learning: Optimal representation geometry and neural- vs dimensional-collapse. *arXiv preprint arXiv:2311.05139*.
- Raphi Kang, Yue Song, Georgia Gkioxari, and Pietro Perona. 2025. Is CLIP ideal? No. can we fix it? Yes! *arXiv preprint arXiv:2503.08723*.

- Chris Dongjoo Kim, Byeongchang Kim, Hyunmin Lee, and Gunhee Kim. 2019. AudioCaps: Generating captions for audios in the wild. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 119–132.
- Yiming Li, Zhifang Guo, Xiangdong Wang, and Hong Liu. 2024. Advancing multi-grained alignment for contrastive language-audio pre-training. In *Proceedings of the 32nd ACM International Conference on Multimedia (ACM MM)*, pages 7356–7365.
- Victor Weixin Liang, Yuhui Zhang, Yongchan Kwon, Serena Yeung, and James Y Zou. 2022. Mind the gap: Understanding the modality gap in multi-modal contrastive representation learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 35, pages 17612–17625.
- Xinhao Mei, Chutong Meng, Haohe Liu, Qiuqiang Kong, Tom Ko, Chengqi Zhao, Mark D Plumbley, Yuexian Zou, and Wenwu Wang. 2024. WavCaps: A ChatGPT-assisted weakly-labelled audio captioning dataset for audio-language multimodal research. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*.
- Microsoft. 2023. [MSCLAP](#). Accessed: 2025.
- Daisuke Niizumi, Daiki Takeuchi, Masahiro Yasuda, Binh Thien Nguyen, Yasunori Ohishi, and Noboru Harada. 2025. M2D-CLAP: Exploring general-purpose audio-language representations beyond CLAP. *IEEE Access*.
- Gustavo Penha, Arthur Câmara, and Claudia Hauff. 2022. Evaluating the robustness of retrieval pipelines with query variation generators. In *Advances in Information Retrieval: 44th European Conference on IR Research (ECIR)*, pages 397–412. Springer.
- Joshua Robinson, Ching-Yao Chuang, Suvrit Sra, and Stefanie Jegelka. 2021. Contrastive learning with hard negative samples. In *The Ninth International Conference on Learning Representations (ICLR)*.
- Ramaneswaran Selvakumar, Sonal Kumar, Hemant Kumar Giri, Nishit Anand, Ashish Seth, Sreyan Ghosh, and Dinesh Manocha. 2025. Do audio-language models understand linguistic variations? In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*.
- Benno Weck and Frederic Font. 2024. The language of sound search: Examining user queries in audio search engines. In *Proceedings of the 9th Workshop on Detection and Classification of Acoustic Scenes and Events (DCASE)*.
- Yusong Wu, Ke Chen, Tianyu Zhang, Yuchen Hui, Taylor Berg-Kirkpatrick, and Shlomo Dubnov. 2023. Large-scale contrastive language-audio pretraining with feature fusion and keyword-to-caption augmentation. In *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 1–5. IEEE.
- HaeJun Yoo, Yongseop Shin, Insung Lee, Myoung-Wan Koo, and Du-Seong Chang. 2026. [Omni-embed-audio: Leveraging multimodal llms for robust audio-text retrieval](#). Preprint, arXiv:2604.18360.
- Yi Yuan, Zhuo Chen, Xubo Liu, Haohe Liu, Xuenan Xu, Dongya Jia, Yuezhe Chen, Mark D Plumbley, and Wenwu Wang. 2024. T-CLAP: Temporal-enhanced contrastive language-audio pretraining. *arXiv preprint arXiv:2404.17806*.
- Jiayi Zhang, Simon Yu, Derek Chong, Anthony Sicilia, Michael R Tomz, Christopher D Manning, and Weiyan Shi. 2025. Verbalized sampling: How to mitigate mode collapse and unlock llm diversity. *arXiv preprint arXiv:2510.01171*.

A CORA Details

A.1 CORA taxonomy

- **Command Query:** Imperative queries that directly request detection or retrieval of a target sound (e.g., “*Detect a dog barking while vehicles are approaching and then driving past on a nearby road.*”)
- **Question Query:** Natural language questions asking whether a specific sound event exists (e.g., “*Are there sounds of a dog barking while vehicles approach and then drive past on a nearby road?*”)
- **Indirect Query:** Indirect or polite request-style queries commonly used in conversational interactions (e.g., “*At your convenience, could you help locate the sound where a dog is barking while vehicles are approaching and then driving past on a nearby road?*”)
- **Key phrase Query:** Keyword-centric queries with minimal sentence structure (e.g., “*dog barking with approaching and passing vehicles on nearby road*”)
- **Statement Query:** Declarative queries that objectively describe sound events without explicit requests or emotional expressions (e.g., “*A dog is barking while vehicles approach and then drive past on a nearby road.*”)

A.2 CORA Generation

CORA was generated using GPT-4.1 with temperature = 1.0, top-p=0.9, and maximum generation length of 2048 tokens. For each query type, we designed different personas and generation constraints to reflect the diversity of real-world user search expressions. We construct CORA from the test and validation splits of four audio-text retrieval datasets: Clotho (1,045 test and 1,045 validation samples), AudioCaps (974 test and 495 validation samples), MECAT (848 test and 160 validation samples), and MACS (500 test and 500 validation samples), resulting in a total of 5,567 source captions.

The 5,567 source captions include 3,367 test and 2,200 validation captions. Retrieval evaluation uses only the test split. The main retrieval analysis uses only test items with all five non-empty query forms. This removes 11 AudioCaps items with an empty indirect query and 51 Clotho items with an empty

key-phrase query. The resulting 3,305 items form the complete evaluation set.

During the generation of indirect, command, and question queries, we observed that query openings often converged to repetitive patterns. To alleviate this issue, CORA introduces a training-free diversification prompting strategy.

The motivation for this approach is based on the observation that LLM generation can be viewed as probabilistic sampling conditioned on prompts. Prior work (Zhang et al., 2025) suggests that prompts can induce different response distributions. We hypothesized that frequently repeated opening expressions correspond to high-probability regions of the generation distribution. Based on this intuition, we first force the model to generate multiple plausible opening expressions and then explicitly prohibit the final query from using any of the generated candidates. This process encourages the model to avoid repeatedly generated prefixes and increases formulation diversity.

Specifically, before generating the final query, the model is first instructed to generate multiple plausible opening expressions for the target query type. The model is then required to select a final opening expression that does not appear in the generated candidate list. By forcing the final query to begin with a non-listed expression, the generation process reduces prefix collapse and increases formulation diversity across query types.

The prompts used for each query type are provided in Appendix N.1.

A.3 CORA Verification

A.3.1 Query Quality Verification

LLM-as-a-Judge and Disjoint Human Verification. We validate CORA queries along two axes: *semantic alignment* (whether the original caption’s meaning is preserved after reformulation) and *query-type validity* (whether each query correctly reflects its intended type). We evaluate 100 samples from each dataset (AudioCaps, Clotho, MACS, and MECAT), resulting in a total of 500 evaluated samples, using two LLM judges—Gemini 2.5 Flash and Qwen2.5-72B-Instruct—both of which exhibit human-like agreement patterns rather than super-consistent behavior (Han et al., 2025), making them well-suited for subjective judgment tasks. Each judge is run three times, showing stable overall performance with variation within 0.3 percentage points across runs. Human verification is addition-

ally conducted by five undergraduate annotators with high English proficiency.

As shown in Table 7, all query types achieve high scores on both axes across all datasets, with most types exceeding 90%. Human verification yields 98.3% on semantic alignment and 97.7% on query-type validity, consistent with the automated results. These findings confirm that CORA queries are semantically faithful and linguistically well-formed with respect to their target types.

For human verification, we recruit 5 undergraduate students with high English proficiency as annotators. All annotators participated as volunteers without compensation and were informed of the research purpose before participation. Each annotator is assigned to a disjoint subset of samples to independently assess semantic alignment and query type validity.

Overlapping Validation and Inter-Annotator Reliability. To address potential individual annotator bias and evaluate inter-annotator agreement (Cohen’s κ), we conducted an additional overlapping human validation on 100 CORA reformulation pairs. All five annotators independently assessed the exact same 100 pairs across both Task 1 (Semantic Alignment) and Task 2 (Query-Type Validity) using the identical rubric provided to the LLM judges.

As summarized in Table 8, annotators demonstrated strong inter-annotator reliability, achieving a mean pairwise Cohen’s κ of 0.970 (range: 0.925–1.000) with 98.8% accuracy for Task 2, and 98.2% raw agreement for Task 1.

Out of 500 judgments in Task 2, only 6 single-annotator errors occurred. As detailed in Table 9, these discrepancies were mostly subtle boundary cases between interrogative/indirect forms and key phrases vs. imperatives.

A.3.2 Five-Form Taxonomy Validation

To examine whether our findings depend on any particular query form, we conduct a leave-one-form-out analysis. Specifically, we remove each of the five query forms and recompute the correlations between RankDrop and TBMD, TAL, and Δ_{move} using the remaining four forms. Across all five leave-one-form-out conditions, both TBMD ($r = 0.607\text{--}0.625$) and TAL ($r = 0.494\text{--}0.526$) remain substantially more strongly correlated with RankDrop than Δ_{move} ($r = 0.060\text{--}0.098$). Thus, the TAL/TBMD $\gg \Delta_{\text{move}}$ pattern is preserved re-

gardless of which query form is excluded, indicating that our main diagnostic conclusion is not driven by any single query form.

A.3.3 Real-World Query Coverage

To examine whether our five query forms reflect how users express information needs in practice, we additionally analyze WildClaims, a dataset of real conversational user interactions. We focus on its *Information seeking* subset, which contains queries where users ask for specific information or facts and thus provides a practical proxy for user queries. We classify all 2,538 information-seeking queries into our five forms (*Question*, *Command*, *Key Phrase*, *Statement*, and *Indirect*) plus a *None* category using Gemini 2.5 Flash with deterministic decoding ($T = 0$).

Overall, 97.6% of the real-user queries are covered by our five forms: 55.7% are Questions, 23.5% Commands, 9.6% Key Phrases, 5.2% Statements, and 3.6% Indirect queries. Only 2.4% are classified as *None*. Most of these cases do not exhibit a clear or typical query form, including copied search-result JSON, webpage dumps, or incomplete fragments. Qualitative examples in Table 10 further illustrate that the same five query forms occur in real-user queries. The classification prompt is provided in Appendix N.2.

These results provide empirical evidence that the five-form taxonomy covers most real-world information-seeking queries. We do not aim to reproduce their natural frequency distribution. Instead, CORA uses balanced reformulations to isolate the effect of query form on retrieval behavior.

A.4 Additional Diversity Analysis

A.4.1 Experimental Setup

To analyze whether CORA increases query diversity in practice, we compare UIQ, CORA(w/o Prompt), and CORA.

UIQ is the original query benchmark. CORA(w/o Prompt) uses the same taxonomy as CORA but removes the diversification strategy. CORA is the full version proposed in this work.

For fair comparison, we sampled 100 examples each from AudioCaps, Clotho, and MECAT, resulting in approximately 300 queries per query type. MACS was excluded because it is not included in UIQ.

Sentence embeddings were extracted using sentence-transformers/all-MiniLM-L6-v2.

Query Type	Human		Gemini 2.5 Flash		Qwen2.5-72B	
	T1	T2	T1	T2	T1	T2
Key Phrase	97.6	97.8	89.7	100.0	100.0	100.0
Statement	97.0	100.0	100.0	100.0	100.0	100.0
Question	99.4	94.9	92.0	97.7	100.0	90.3
Command	98.2	100.0	93.3	95.0	95.3	96.0
Indirect	99.4	95.6	99.0	100.0	100.0	100.0
Overall	98.3	97.7	94.8	98.5	99.1	97.3

Table 7: Verification accuracy (%) for semantic alignment (Task 1) and query-type validity (Task 2) across five CORA query types.

Metric / Evaluation Axis	Value
Task 1 Raw Agreement (Semantic Alignment)	98.2%
Task 2 Accuracy (Query-Type Validity)	98.8%
Mean Pairwise Cohen’s κ	0.970
Pairwise Cohen’s κ Range	0.925–1.000

Table 8: Summary of additional overlapping human validation across 5 annotators on the same 100 pairs.

Error Type	Confusion Pair	Count
Interrogative form	Question $\leftrightarrow$ Indirect	3
Phrase vs. sentence	Key phrase $\leftrightarrow$ Statement	1
Phrase vs. imperative	Key phrase $\leftrightarrow$ Command	1
Declarative vs. indirect	Statement $\leftrightarrow$ Indirect	1
Total		6

Table 9: Breakdown of query-type confusion errors observed in the overlapping human validation.

A.4.2 Metrics

We measure semantic diversity using a pairwise cosine-distance-based diversity metric. Global diversity is computed as the average cosine distance between all query embeddings:

$$D_{\text{global}} = \text{mean}_{i < j} \left(1 - e(q_i)^\top e(q_j) \right)$$

where i and j index different queries, and $e(q_i)$ denotes the normalized sentence embedding of query q_i . Higher global diversity indicates that query formulations are more widely distributed in the embedding space.

To additionally measure formulation diversity within the same query type, we compute within-type diversity:

$$D_{\text{within}} = |T|^{-1} \sum_{t \in T} \text{mean}_{i < j} \left(1 - e(q_i^t)^\top e(q_j^t) \right)$$

where q_i^t and q_j^t denote two different queries belonging to query type t . Since the diversification

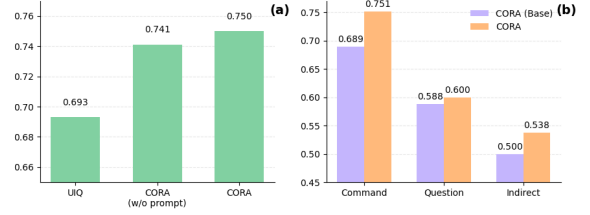


Figure 4: Semantic diversity comparison between UIQ, CORA(w/o Prompt), and CORA. (a) Global semantic diversity across all query embeddings. (b) Within-type semantic diversity for command, question, and indirect queries. CORA consistently improves diversity over CORA(w/o Prompt), with the largest gain observed for the command type.

prompting strategy is applied only to command, indirect, and question queries, within-type diversity is evaluated exclusively on these query types.

A.4.3 UIQ vs. CORA vs. CORA(w/o Prompt)

Compared to UIQ, CORA(w/o Prompt) achieves substantially higher global semantic diversity (0.693 $\rightarrow$ 0.741; Figure 4(a)). This suggests that introducing indirect query types expands the overall embedding distribution by incorporating conversational and request-oriented formulations that occupy regions distinct from existing query styles.

CORA further improves global diversity over CORA(w/o Prompt) (0.741 $\rightarrow$ 0.750; Figure 4(a)). Since both variants share the same query taxonomy, the improvement mainly comes from the proposed diversification prompting strategy rather than query-type expansion itself.

Within-type analysis further shows that the command type benefits the most from the diversification strategy (0.689 $\rightarrow$ 0.751; Figure 4(b)). In contrast, question queries exhibit relatively smaller gains because yes/no question structures naturally constrain formulation variability.

Form	Real-user query (WildClaims)	CORA query
Key Phrase	Free resources to get AMHS based messages	Crowd speech laughter
Statement	The bomb was dropped on Hiroshima at around 8 am.	Many people are speaking and laughing.
Question	Can you find leet cheatsheet?	Does the recording feature pigeons cooing, bird wings flapping, gravel shuffling, and wood clacking?
Command	Show me sources behind all of these bullets.	Retrieve speech and laughter from many people.
Indirect	Can you please provide me with citable sources of journals that discusses the topic of organizational innovation technology adoption and the inconsistencies between research studies.	Would you be able to help me locate the audio where a train is running on train tracks and a steam engine horn is whistling?

Table 10: Examples of real-user information-seeking queries from WildClaims and CORA queries exhibiting the same query forms.

Query set	Forms	Evaluations	MeanRD
CORA	5	56,100	+5.28
CORA w/o div.	5	56,100	+4.17
Released UIQ	4	44,880	−5.78

Table 11: MeanRD for three query sets evaluated on the same 2,805 items and per-dataset candidate pools.

Query set (N)	Δ_{move}	TAL	Dynamic-HN TBMD
CORA (56,100)	.092/.071	.457/.574	.535/.667
CORA w/o diversification (56,100)	.081/.064	.443/.578	.519/.660
Released UIQ (44,880)	−.075/ − .073	.508/.669	.538/.736

Table 12: Pearson r /Spearman ρ correlations with RankDrop under matched per-dataset candidate pools. N counts item-by-query-form-by-retriever observations across 2,805 items and four retrievers. This is a descriptive comparison because query generation, grounding, and query-type inventories differ.

A.5 Same-Pool Retrieval Comparison

We compare CORA, CORA without diversification, and released UIQ on the same 2,805 test items: 963 from AudioCaps, 994 from Clotho, and 848 from MECAT. MACS is not included in UIQ. The evaluation uses the same source captions, target audios, per-dataset candidate pools, and four CLAP retrievers.

The paired MeanRD difference is +1.10 for CORA versus CORA without diversification (95% cluster-bootstrap CI [0.75, 1.45]) and +11.06 for CORA versus UIQ ([9.35, 12.82]). These are descriptive comparisons because the query sets differ in generation conditions and, for UIQ, query types. The generator also differs: CORA and CORA without diversification were generated with GPT-4.1, whereas UIQ was generated with GPT-5.1. All three query sets preserve the TAL/TBMD-over- Δ_{move} ordering, so the pattern is not confined to GPT-4.1-generated queries, although this is not a

controlled generator-stability experiment.

B Experimental Setup Details

Datasets. We evaluate retrieval robustness on AudioCaps, Clotho, MACS, and MECAT. AudioCaps is based on AudioSet, and several clips overlap with the WavCaps AudioSet_SL subset. Similarly, Clotho is built on Freesound, which substantially overlaps with the WavCaps Freesound subset. Such overlap may lead to overestimated retrieval performance. MECAT is constructed from ACAV100M web-video audio data and does not directly overlap with WavCaps sources. MACS is based on TAU Urban Acoustic Scenes 2019 and consists of real-world urban scene recordings. Therefore, MECAT and MACS can be regarded as relatively leakage-free benchmarks.

Retrieval Models. We evaluate four CLAP-family retrievers without additional fine-tuning:

- **LAION-CLAP:** a representative zero-shot T2A retrieval model based on contrastive audio-language pretraining.
- **M2D-CLAP:** a retrieval model utilizing masked modeling-based representation learning.
- **MGA-CLAP:** a retrieval model designed for multi-granularity audio-text alignment.
- **MS-CLAP:** a CLAP variant trained with a multi-stage training strategy.

C Retriever-Level CORA Evaluation

Table 13 provides the global marginal retrieval performance and paired rank degradation (MeanRD) across all evaluated retrievers, including both CLAP models and external robustness-oriented

models. As discussed in Section 4.2, analyzing aggregate $\Delta R@5$ alone can be misleading because it obscures instance-level target retention. For example, while MS-CLAP shows a positive aggregate $\Delta R@5$ (+3.07%), this increase starts from a very low source caption baseline (17.37%); its MeanRD remains near zero, indicating that the original targets are not necessarily better preserved. Similarly, external models like RobustCLAP show severe paired rank drops (MeanRD 9.408) despite their robustness-oriented training, which aggregate metrics might partially mask. This comprehensive retriever-level summary supports our methodological choice to rely on paired, continuous RD metrics rather than aggregate top- k recall changes.

D Full Dataset-by-Model-by-Form Retrieval Results

Section 4.2 reports marginal summaries along the dataset form and dataset model axes. Those summaries are compact, but they average over one axis of variation. For completeness, Table 14 reports the full dataset,model,query-form retrieval breakdown. The table follows the main outcome flow and includes only aggregate Recall@5 and paired MeanRD.

The full breakdown reinforces the main conclusion that CORA sensitivity is interactional. A query form that is stable for one dataset-model pair can be harmful for another, and aggregate $\Delta R@5$ should be interpreted together with paired MeanRD.

E Interaction Analysis for Query-Form Sensitivity

Section 4.2 shows that CORA sensitivity varies along both dataset-form and dataset-model axes. We further test whether this pattern is better described by interactional effects rather than by additive model, dataset, and query-form effects alone.

We predict rank degradation using grouped cross-validation by audio ID. This split prevents CORA pairs derived from the same target audio from appearing in both the training and validation folds. The prediction target is $RD(p)$, where positive values indicate that the target rank becomes worse after reformulation.

Table 18 reports prediction performance under increasingly specific feature sets. Additive model, dataset, and query-form terms provide limited predictive power. Adding model-dataset and query in-

teraction terms improves performance, and the full model-dataset-query cell gives the highest correlation. This supports the main-text conclusion that query-form sensitivity is conditional on the model-dataset regime rather than being a universal property of the surface query form.

F Hard Negative Definition Sensitivity

Section 3.4 defines TBMD using the strongest non-target audio for each query. Specifically, the source query uses h_{src} , the top-1 non-target audio under q_{src} , and the CORA query uses h_{cora} , the top-1 non-target audio under q_{cora} . This dynamic top-1 definition is the main diagnostic used in the paper:

$$TBMD_{dyn} = [\cos(q_{src}, a_{tgt}) - \cos(q_{src}, h_{src})] - [\cos(q_{cora}, a_{tgt}) - \cos(q_{cora}, h_{cora})]. \quad (8)$$

We include a sensitivity check to verify that the boundary signal is not an artifact of this particular hard negative anchoring choice. We compare the main dynamic top-1 definition against two fixed-top-1 alternatives. The first fixes the hard negative to the strongest CORA-query non-target audio and evaluates both query forms against that same competitor:

$$TBMD_{CORA-HN} = [\cos(q_{src}, a_{tgt}) - \cos(q_{src}, h_{cora})] - [\cos(q_{cora}, a_{tgt}) - \cos(q_{cora}, h_{cora})]. \quad (9)$$

The second fixes the hard negative to the strongest source-query non-target audio and evaluates both query forms against that source-side competitor:

$$TBMD_{src-HN} = [\cos(q_{src}, a_{tgt}) - \cos(q_{src}, h_{src})] - [\cos(q_{cora}, a_{tgt}) - \cos(q_{cora}, h_{src})]. \quad (10)$$

All three variants are degradation-oriented, so larger values indicate a larger loss of target margin after reformulation.

Table 19 shows that all top-1 boundary variants are positively associated with RD. The fixed-source variant is weaker because a source-side hard negative can miss a new competitor that becomes salient only after reformulation. The fixed-CORA variant is stronger, but it evaluates the source and CORA queries against a boundary selected only after reformulation. The main dynamic top-1 definition is strongest because it compares each query against its own most competitive non-target audio. This supports the use of dynamic top-1 TBMD as the primary boundary diagnostic.

Retriever	Type	Source R@5	CORA R@5	$\Delta R@5$	MeanRD
LAION	single	29.68	22.17	-7.51	104.71
MS-CLAP	single	17.37	20.44	3.07	-2.67
M2D	single	34.89	29.57	-5.31	2.46
MGA	single	34.52	31.69	-2.83	30.34
MultiCLAP-ScoreAvg-Z	fusion	38.67	35.75	-2.92	24.29
MultiCLAP-ScoreAvg-MinMax	fusion	38.28	35.50	-2.77	25.04
OEA-Qwen3B-Cl	external	44.01	43.34	-0.67	0.603
OEA-Qwen3B-AC	external	42.04	41.17	-0.87	0.479
RobustCLAP	external	47.24	35.47	-11.77	9.408

Table 13: Comprehensive retriever-level performance under the CORA protocol. R@5 and $\Delta R@5$ values are reported as percentages. MeanRD is computed over all CORA pairs. This table illustrates the limitation of aggregate $\Delta R@5$ and motivates the use of paired MeanRD to accurately capture instance-level rank degradation.

Model	Query form	Source R@5	CORA R@5	$\Delta R@5$	MeanRD
AudioCaps					
LAION	Statement	54.1	40.2	-13.9	32.16
	Question	54.1	38.4	-15.7	62.50
	Key phrase	54.1	37.9	-16.2	58.76
	Command	54.1	38.2	-15.9	47.29
	Indirect	54.1	41.8	-12.3	47.43
M2D	Statement	68.7	56.7	-12.0	21.63
	Question	68.7	59.3	-9.4	13.53
	Key phrase	68.7	64.8	-3.9	0.24
	Command	68.7	43.5	-25.2	59.77
	Indirect	68.7	53.8	-15.0	18.79
MGA	Statement	63.2	61.7	-1.6	4.28
	Question	63.2	55.6	-7.7	19.69
	Key phrase	63.2	61.7	-1.6	2.62
	Command	63.2	59.4	-3.8	13.13
	Indirect	63.2	35.9	-27.3	115.80
MS-CLAP	Statement	10.1	37.0	26.9	-143.23
	Question	10.1	39.0	29.0	-151.06
	Key phrase	10.1	9.0	-1.0	32.72
	Command	10.1	39.0	29.0	-152.71
	Indirect	10.1	34.1	24.0	-135.51

Table 14: Full dataset-model-query-form retrieval results under CORA reformulation (AudioCaps). R@5 and $\Delta R@5$ are percentage points; MeanRD is $r_{\text{CORA}} - r_{\text{src}}$.

G Anchor and Content Controls for Target Alignment Loss

The main diagnostic analysis uses source queries as anchors. This choice is central to the CORA protocol: the source query defines the original benchmark retrieval input, and CORA measures how the same target audio changes rank when that query is reformulated into user style CORA queries. However, this design also raises a possible concern. The observed Target Alignment Loss (TAL) may be tied to the particular source-query anchor, or to lexical and content-realization differences between source queries and CORA queries. We therefore include two supporting controls that modify the anchoring condition.

First, in the statement-anchor control, we use the Statement query as the anchor and compare the remaining four CORA query forms against it. We denote this setting as CORA4. Because the Statement query is itself a CORA query and is closest in form to a descriptive retrieval query, this control reduces dependence on the original source query while pre-

serving a paired reformulation setting. Second, in the content-fixed wrapper control, the core acoustic content is held fixed and only the surrounding query wrapper is varied. This control tests whether the TAL signal persists when the main variation comes from query formulation rather than changes in the described sound content.

These controls are not intended to replace the main source-query-anchored analysis, because the primary CORA setting is precisely the shift from source queries to user style CORA queries. Instead, they serve as robustness checks for whether the TAL-RD association is unique to the source-query comparison.

Table 20 shows that the TAL-RD association becomes weaker when the original source-query anchor is removed or when the acoustic content is fixed. This reduction is expected, since these controls no longer exactly match the main CORA protocol. Nevertheless, the association remains consistently positive. The result suggests that the target-alignment diagnostic is not solely an artifact of using source queries as anchors, nor solely a

Model	Query form	Source R@5	CORA R@5	$\Delta R@5$	MeanRD
Clotho					
LAION	Statement	29.9	32.9	3.0	-14.36
	Question	29.9	25.6	-4.3	2.69
	Key phrase	29.9	13.9	-16.0	164.85
	Command	29.9	27.7	-2.2	-13.62
	Indirect	29.9	23.3	-6.5	20.05
M2D	Statement	37.9	38.1	0.2	-1.75
	Question	37.9	32.8	-5.1	5.22
	Key phrase	37.9	21.1	-16.8	49.86
	Command	37.9	36.2	-1.7	1.70
	Indirect	37.9	27.8	-10.2	29.95
MGA	Statement	37.8	37.0	-0.8	0.10
	Question	37.8	35.9	-1.9	18.08
	Key phrase	37.8	36.0	-1.8	5.61
	Command	37.8	35.6	-2.2	17.78
	Indirect	37.8	33.8	-4.0	40.26
MS-CLAP	Statement	34.0	35.3	1.3	-9.94
	Question	34.0	30.9	-3.1	2.06
	Key phrase	34.0	7.7	-26.3	239.36
	Command	34.0	35.1	1.1	-10.13
	Indirect	34.0	30.5	-3.5	6.51

Table 15: Full dataset–model–query-form retrieval results under CORA reformulation (Clotho). R@5 and $\Delta R@5$ are percentage points; MeanRD is $r_{\text{CORA}} - r_{\text{src}}$.

Model	Query form	Source R@5	CORA R@5	$\Delta R@5$	MeanRD
MACS					
LAION	Statement	16.2	0.8	-15.4	361.91
	Question	16.2	4.2	-12.0	299.02
	Key phrase	16.2	16.2	0.0	2.15
	Command	16.2	8.8	-7.4	166.96
	Indirect	16.2	10.4	-5.8	243.89
M2D	Statement	1.4	5.8	4.4	-169.65
	Question	1.4	3.4	2.0	-141.18
	Key phrase	1.4	1.6	0.2	-50.01
	Command	1.4	7.8	6.4	-206.73
	Indirect	1.4	1.2	-0.2	64.83
MGA	Statement	8.6	8.0	-0.6	5.72
	Question	8.6	14.2	5.6	-34.27
	Key phrase	8.6	9.0	0.4	5.56
	Command	8.6	7.6	-1.0	5.67
	Indirect	8.6	14.2	5.6	-23.43
MS-CLAP	Statement	14.0	4.6	-9.4	110.13
	Question	14.0	9.2	-4.8	63.43
	Key phrase	14.0	12.4	-1.6	22.37
	Command	14.0	6.4	-7.6	92.76
	Indirect	14.0	5.4	-8.6	74.67

Table 16: Full dataset–model–query-form retrieval results under CORA reformulation (MACS). R@5 and $\Delta R@5$ are percentage points; MeanRD is $r_{\text{CORA}} - r_{\text{src}}$.

consequence of source-query-to-CORA-query content mismatch. We therefore treat these results as supporting robustness checks for the main source-query-anchored analysis.

H TBMD Control Analysis

Because TBMD is computed from the same similarity scores that determine rank, its correlation with RD in Section 4.3.1 could partly reflect this construction rather than a genuine boundary effect. We test whether TBMD adds explanatory power over TAL once we control for dataset, retrieval model, query form, source rank, and candidate pool size.

For both the per-dataset and pooled cross-dataset settings (66,100 pairs each), we fit a 5-fold grouped cross-validated Ridge regression (Appendix K.3) predicting RD, first from TAL alone and then from TAL together with dynamic-HN TBMD, both with fixed effects for dataset, model, and query form. We also compute the partial correlation of TBMD with RD after residualizing out TAL, dataset, model,

query form, Δ_{move} , source rank, and pool size via OLS.

TBMD improves prediction over TAL alone in both settings, and its partial correlation with RD stays positive once these confounds are held fixed, though the partial association is smaller in the pooled setting. We therefore treat TBMD as a diagnostic signal that adds information beyond TAL, rather than evidence that TBMD alone drives rank degradation.

I Boundary-Margin Token Saliency Audit

We further examine whether the boundary-centered RD pattern is visible at the token-attribution level. This analysis is intended as a local diagnostic audit rather than a causal token-deletion experiment. For each unchanged CORA query, we compute token-level Gradient $\times$ Input saliency with respect to the target boundary margin.

We evaluate the four CLAP retrievers used in the main experiments: LAION-CLAP, MS-CLAP,

Model	Query form	Source R@5	CORA R@5	$\Delta R@5$	MeanRD
MECAT					
LAION	Statement	9.7	7.0	-2.7	87.86
	Question	9.7	10.4	0.7	161.31
	Key phrase	9.7	3.8	-5.9	323.46
	Command	9.7	9.9	0.2	132.71
	Indirect	9.7	9.4	-0.2	233.13
M2D	Statement	12.6	12.7	0.1	8.76
	Question	12.6	14.0	1.4	5.68
	Key phrase	12.6	15.9	3.3	-113.27
	Command	12.6	9.3	-3.3	166.85
	Indirect	12.6	13.9	1.3	-11.71
MGA	Statement	13.3	14.4	1.1	-0.51
	Question	13.3	12.5	-0.8	93.64
	Key phrase	13.3	14.6	1.3	18.39
	Command	13.3	12.4	-0.9	73.97
	Indirect	13.3	11.9	-1.4	157.28
MS-CLAP	Statement	8.1	5.3	-2.8	-53.57
	Question	8.1	2.7	-5.4	143.58
	Key phrase	8.1	19.3	11.2	-386.79
	Command	8.1	3.2	-5.0	251.05
	Indirect	8.1	2.1	-6.0	136.80

Table 17: Full dataset–model–query-form retrieval results under CORA reformulation (MECAT). R@5 and $\Delta R@5$ are percentage points; MeanRD is $r_{\text{CORA}} - r_{\text{src}}$.

Feature terms	Pearson r	Spearman ρ
additive model/dataset/query	0.240	0.244
model-dataset + query	0.285	0.257
pairwise query interactions	0.309	0.317
full model-dataset-query cell	0.376	0.391

Table 18: Grouped cross-validated prediction of RD. Interaction terms improve prediction over additive model, dataset, and query form terms, indicating that CORA sensitivity is conditional on the model-dataset regime.

Top-1 boundary degradation definition	Pearson w/ RD
CORA-HN fixed TBMD	0.487
Source-HN fixed TBMD	0.317
Dynamic top-1 TBMD (pooled)	0.598

Table 19: Sensitivity of TBMD to top-1 hard negative anchoring choices. All variants use top-1 non-target audios, matching the main diagnostic setting. The dynamic top-1 definition used in the main text gives the strongest association with RD.

M2D-CLAP, and MGA-CLAP. The analysis is restricted to sentence-like CORA forms: Statement, Question, Command, and Indirect. We exclude Key phrase queries because compressed phrase queries do not share the same function-word and intent-word structure as the other query forms. For each model, we sample up to 10 examples per dataset, query form, and RD group, yielding 320 query pairs per model and 1,280 query pairs in total. The two RD groups are:

$$\text{rank-worsened} : r_{\text{CORA}} > r_{\text{src}},$$

$$\text{rank-improved/stable} : r_{\text{CORA}} \leq r_{\text{src}}.$$

For a CORA query q , let a_{tgt} be the target audio and let a_{hn} be the highest-scoring non-target audio

Anchor/control	N	r(TAL, RD)
Source query $\rightarrow$ CORA query	66,100	0.615
Statement query $\rightarrow$ CORA4	45,888	0.399
Content-fixed wrapper control	32,320	0.406

Table 20: Anchor and content controls for Target Alignment Loss (TAL), computed with the pooled cross-dataset candidate set. TAL is degradation-oriented, so positive correlations indicate that larger loss of target alignment is associated with larger RD.

Analysis	Per-dataset	Pooled
TAL-only (Pearson r)	0.561	0.615
TAL + dynamic-HN TBMD	0.644	0.626
Increment	+0.083	+0.010
Partial r beyond TAL	0.382	0.144

Table 21: Control analysis for TBMD. Adding dynamic-HN TBMD to TAL improves RD prediction in both settings, and TBMD keeps a positive partial correlation with RD once TAL, dataset, model, query form, Δ_{move} , source rank, and pool size are held fixed.

under the CORA query. The local boundary margin is

$$M(q) = \cos(q, a_{\text{tgt}}) - \cos(q, a_{\text{hn}}).$$

A positive margin means that the query scores the target above the local hard negative, while a negative margin means that the hard negative is favored over the target. For each token i , with input embedding e_i , we compute Gradient $\times$ Input saliency with respect to this margin:

$$s_i = e_i \cdot \frac{\partial M(q)}{\partial e_i}.$$

We normalize token saliency within each query by

total absolute saliency:

$$\tilde{s}_i = \frac{s_i}{\sum_j |s_j| + \epsilon}.$$

Positive saliency indicates local support for the target over the hard negative, while negative saliency indicates local support for the hard negative over the target.

Tokens are grouped into three surface categories:

- **acoustic content:** tokens describing sound events, sources, acoustic attributes, or scenes;
- **query intent:** tokens such as *find*, *search*, *retrieve*, *audio*, and *sound*;
- **function/form:** determiners, auxiliaries, prepositions, question markers, politeness markers, and other form-related tokens.

For each group, we separately aggregate positive and negative saliency mass. For example, Content+ measures positive boundary-margin support assigned to acoustic-content tokens, while Form- measures negative boundary-margin contribution assigned to function/form tokens.

Table 22 shows that the clearest group-level separation is the boundary margin itself. Across all four models, rank-worsened pairs have lower mean target boundary margins than rank-improved or stable pairs. For LAION, the mean margin decreases from -0.0928 to -0.2189 ; for MS-CLAP, from -0.0697 to -0.1250 ; for M2D, from -0.0169 to -0.0430 ; and for MGA, from -0.0459 to -0.1138 . This is consistent with the main TBMD result: harmful reformulations are associated with weaker local target-boundary margins.

However, the token-group shares themselves do not provide a stable explanation of RD. Content+, Content-, Form-, and Intent- vary only modestly between the two RD groups, and no token category separates rank-worsened pairs consistently across all models. For example, Form- is slightly higher for rank-worsened pairs in LAION and MGA, but lower in MS-CLAP and M2D. This suggests that RD is not reducible to a single harmful token class such as function words or intent words.

We next test whether token-group saliency features improve RD prediction beyond the boundary diagnostics used in the main text. We use grouped cross-validation by audio ID and predict continuous RD. The feature sets are: fixed effects only, query movement, token-group saliency, boundary

features, and boundary features plus token-group saliency. Boundary features include query movement, target-alignment loss, TBMD, and boundary projection.

Table 23 shows that token-group saliency alone is weak for continuous RD prediction, reaching only Pearson $r = 0.090$ and Spearman $\rho = 0.037$ in the pooled setting. Query movement is also weak, with Pearson $r = 0.115$ and Spearman $\rho = 0.045$. By contrast, boundary features reach Pearson $r = 0.634$ and Spearman $\rho = 0.591$. Adding token-group saliency to the boundary features does not improve prediction; performance changes from $r = 0.634$, $\rho = 0.591$ to $r = 0.632$, $\rho = 0.581$.

This audit gives a limited but useful role to token-level attribution. Token saliency confirms that rank-worsened queries have poorer local target boundary margins, supporting the boundary-centered interpretation. At the same time, coarse token groups do not explain RD beyond the boundary diagnostics. We therefore treat token saliency as a descriptive local attribution tool rather than evidence that a particular token class causally drives CORA degradation.

J Dataset-Level Diagnostic Correlations

Section 4.3.1 reports the pooled diagnostic correlations between RD and the three embedding-space signals. Here, we provide the dataset-level breakdown to verify that the main pattern is not driven by a single dataset. For each dataset, we compute Pearson correlations between RD and query movement, Target Alignment Loss (TAL), and Target Boundary Margin Degradation (TBMD).

Table 24 shows that the main diagnostic pattern is not specific to a single dataset. Query movement has weak or variable association with RD, ranging from near zero on AudioCaps to moderate association on MACS. In contrast, TAL and TBMD remain positively associated with RD across all four datasets. This supports the interpretation that CORA degradation is better characterized by target-alignment loss and target-boundary margin degradation than by raw movement away from the source query.

K Projection Analysis Details

This appendix provides the formal details for the projection analysis in the main text. The goal of this analysis is to test whether CORA-induced RD is better explained by the *direction* of the source-

Model	RankDrop group	N	MeanRD	Content+	Content-	Form-	Intent-	Mean margin
LAION	worsened	160	71.819	0.2832	0.2387	0.1205	0.0521	-0.2189
LAION	improved/stable	160	-37.863	0.2799	0.2647	0.1143	0.0409	-0.0928
MS-CLAP	worsened	160	45.581	0.2774	0.2840	0.1025	0.0486	-0.1250
MS-CLAP	improved/stable	160	-50.213	0.2696	0.2885	0.1191	0.0389	-0.0697
M2D	worsened	160	31.356	0.2780	0.2915	0.1178	0.0404	-0.0430
M2D	improved/stable	160	-23.938	0.3025	0.2710	0.1243	0.0466	-0.0169
MGA	worsened	160	36.269	0.3080	0.3267	0.1009	0.0420	-0.1138
MGA	improved/stable	160	-19.813	0.3166	0.3231	0.0946	0.0452	-0.0459

Table 22: Token-group saliency shares by RD group. Mean margin is the local target boundary margin under the CORA query. Across all four models, rank-worsened pairs have lower mean boundary margins than rank-improved or stable pairs.

Feature set	Pearson r	Spearman ρ
Fixed effects only	0.077	0.007
Query movement	0.115	0.045
Token-group saliency	0.090	0.037
Boundary features	0.634	0.591
Boundary + token saliency	0.632	0.581

Table 23: Grouped cross-validated RD prediction from token saliency and boundary features. Token-group saliency alone has weak predictive value, and adding it to boundary features does not improve prediction.

Dataset	N	MeanRD	Worsened (%)	$r(\Delta_{\text{move}})$	$r(\text{TAL})$	$r(\text{TBMD})$
AudioCaps	19,260	-1.61	43.2	-0.016	0.531	0.547
Clotho	19,880	27.71	51.6	0.221	0.580	0.585
MACS	10,000	49.69	50.2	0.361	0.737	0.716
MECAT	16,960	71.43	55.5	0.216	0.700	0.668

Table 24: Dataset-level diagnostic correlations with RD. Worsened denotes the percentage of pairs for which the CORA-query rank is lower than the source-query rank. TAL and TBMD are consistently more associated with RD than query movement across datasets.

to-CORA query shift than by the shift magnitude alone. All embeddings are ℓ_2 -normalized, and cosine similarity is therefore computed as an inner product. For any non-zero vector x , we write

$$\hat{x} = \frac{x}{\|x\|_2 + \varepsilon}$$

for numerical stability.

K.1 Mathematical Formulation

Given a source caption embedding q_{src} and the corresponding CORA query embedding q_{cora} , we first compute the raw source-to-CORA shift:

$$\delta = q_{\text{cora}} - q_{\text{src}}.$$

Because the retrieval embeddings lie on the unit sphere, we remove the radial component of this

shift and analyze the tangent-space displacement at the source caption anchor:

$$d = \delta - \langle q_{\text{src}}, \delta \rangle q_{\text{src}}.$$

The movement-magnitude feature used in the main projection experiment is

$$\|d\|_2.$$

We intentionally denote this feature as $\|d\|_2$, rather than Δ_{move} , to distinguish it from the cosine-distance query movement metric used in the TBMD diagnostic section.

Target-audio projection. To measure whether the query shift moves toward the target audio, we first project the target audio embedding onto the same tangent space:

$$u_{\text{audio}} = a_{\text{tgt}} - \widehat{\langle q_{\text{src}}, a_{\text{tgt}} \rangle} q_{\text{src}}.$$

The target-audio projection is then

$$p_{\text{audio}} = \langle d, u_{\text{audio}} \rangle.$$

A larger positive value indicates that the CORA shift moves in a direction more aligned with the target audio representation, whereas a negative value indicates movement away from the target direction.

Boundary projection. To measure whether the query shift preserves local separation between the target and a nearby non-target competitor, we define a target boundary direction. Given the source-anchored hard negative h_{src} defined in Appendix K.2, the tangent-space boundary direction is

$$u_{\text{boundary}} = (a_{\text{tgt}} - h_{\text{src}}) - \widehat{\langle q_{\text{src}}, a_{\text{tgt}} - h_{\text{src}} \rangle} q_{\text{src}}.$$

The boundary projection is

$$p_{\text{boundary}} = \langle d, u_{\text{boundary}} \rangle.$$

This feature approximates whether the source-to-CORA query shift moves in a direction that increases or decreases the local target-minus-hard-negative score difference. Positive values indicate movement in a target-favoring boundary direction, while negative values indicate movement toward weakening the local retrieval boundary.

K.2 Hard Negative Selection

The projection experiment uses a source-anchored hard negative so that the analysis tests whether the CORA shift preserves the original local retrieval boundary. This choice is complementary to the dynamic hard negative definition used for TBMD: TBMD measures realized margin degradation under each query, whereas the projection analysis measures the direction of the source-to-CORA shift relative to a source-side boundary.

For each audio item i and CORA form f , we first construct a boundary candidate pool from both retrieval neighborhoods. Let $S_{\text{src}}^5(i)$ be the top-5 non-target audios retrieved by the source query, and let $S_f^5(i)$ be the top-5 non-target audios retrieved by the CORA query of form f . The target audio is excluded from both sets. We define

$$P_{i,f} = S_{\text{src}}^5(i) \cup S_f^5(i).$$

The final hard negative is selected as the highest-scoring non-target audio under the source query within this union pool:

$$h_{\text{src}}^{(i,f)} = \arg \max_{a_j \in P_{i,f}} \cos(q_{\text{src}}^{(i)}, a_j).$$

Using the union pool makes the candidate boundary sensitive to competitors that are salient either before or after reformulation, while selecting the final hard negative with the source query keeps the analyzed boundary anchored to the pre-reformulation retrieval state.

K.3 Regression Details

We conduct the projection analysis on AudioCaps, Clotho, MACS, and MECAT using four CLAP-family retrievers: LAION-CLAP, M2D-CLAP, MGA-CLAP, and MS-CLAP. For each source caption, we evaluate five CORA query forms: Key phrase, Statement, Question, Command, and Indirect. The final evaluation contains 66,100 CORA pairs.

The prediction target is continuous RD,

$$\text{RD} = \text{rank}_{\text{cora}} - \text{rank}_{\text{src}},$$

where larger values indicate that the target audio is ranked lower after reformulation. We use 5-fold grouped cross-validation with groups defined by `dataset::audio_id`, preventing different CORA forms from the same audio sample from appearing in both the training and test folds.

The regression pipeline consists of `DictVectorizer`, `StandardScaler` with `with_mean=False`, and `Ridge` with `alpha=1.0` and `solver=lsqr`.

All settings include fixed effects for dataset, retrieval model, and query form. We then compare the following feature sets:

- fixed effects only;
- fixed effects + $\|d\|_2$;
- fixed effects + p_{audio} ;
- fixed effects + p_{boundary} .

Each projection row adds only the named continuous feature to the same fixed-effect baseline. This isolates whether RD is better explained by movement magnitude, target-oriented direction, or boundary-oriented direction. Prediction quality is measured by Pearson correlation and Spearman correlation between out-of-fold predictions and ground-truth RD values.

K.4 Interpretation

The projection analysis is designed to complement the scalar TAL and TBMD diagnostics in the main text. TAL and TBMD measure realized changes in target alignment and target boundary margin. By contrast, the projection features ask whether the actual embedding displacement from the source caption to the CORA query points in a target-preserving or boundary-preserving direction.

The main results show that adding $\|d\|_2$ gives only a small improvement over fixed effects, while p_{audio} and p_{boundary} provide substantially stronger RD prediction. This supports the directional boundary hypothesis: harmful reformulations are not simply those that move far from the source caption, but those whose embedding shift fails to move toward the target audio or fails to preserve the local target boundary.

Retriever	Source R@5	CORA R@5	$\Delta R@5$	MeanRD
M2D	34.89	29.57	-5.31	2.46
MGA	34.52	31.69	-2.83	30.34
MultiCLAP-ScoreAvg-Z	38.67	35.75	-2.92	24.29
MultiCLAP-ScoreAvg-MinMax	38.28	35.50	-2.77	25.04

Table 25: Additional inference-time score fusion check under the CORA protocol. M2D and MGA are included as strong single-retriever references. R@5 and $\Delta R@5$ values are reported as percentages.

L Additional Inference-Time Fusion Check

We additionally evaluate simple inference-time score fusion baselines that average scores from multiple CLAP retrievers. These baselines are included as supporting checks for aggregate retrieval strength, not as robustness-oriented training methods. As shown in Table 25, score fusion improves aggregate Source R@5 and CORA R@5 over individual CLAP retrievers, but it is not treated as evidence of query-form robustness.

M Query Preprocessing

We examine whether query preprocessing can recover retrieval performance degraded by query reformulation and whether the diagnostic pattern observed in our main analysis persists after preprocessing. Our main experiments evaluate the retrievers directly, without applying any external query rewriting. In contrast, the experiments in this section introduce an external preprocessing module before retrieval (preprocessor $\rightarrow$ retriever).

We evaluate three preprocessing conditions on a balanced sample of 500 source items, with 125 items sampled from each of the four datasets used in our main experiments (Clotho, AudioCaps, MECAT, and MACS). We use all five CORA query forms and four CLAP retrievers, with retrieval performed over the full candidate pool of each dataset.

We consider three conditions: **No canonicalizer**, **General rewriting**, and **CORA-aware rewriting**. In the No canonicalizer condition, each CORA query is directly provided to the retriever. General rewriting uses a generic rewriter that is not given the CORA query-form taxonomy. CORA-aware rewriting, in contrast, is given the query’s CORA form and the definition of that form, allowing the rewriter to use this information when rewriting the query into a caption-style description. However, such form information is not generally available in real-world settings. Therefore, we treat CORA-aware rewriting as a type-aware diagnostic condi-

Condition	RD	R@5	Δ_{move}	TAL	TBMD
No canonicalizer	6.33	37.33	0.103	0.551	0.653
General rewriting	4.14	38.10	0.080	0.519	0.634
CORA-aware	-0.03	40.02	0.020	0.546	0.683

Table 26: Effect of query preprocessing on retrieval performance and diagnostic associations. RD denotes Mean RankDrop, and R@5 is reported in percent. Δ_{move} , TAL, and TBMD report Pearson’s r with RD. Source-query R@5 is 38.95% in all conditions.

tion rather than a directly deployable preprocessing method. The prompts used for both rewriting conditions are provided in Appendix N.3.

As shown in Table 26, query preprocessing reduces average rank degradation. With 95% cluster-bootstrap CIs, General and CORA-aware rewriting reduce MeanRD by 2.18 (CI [0.27, 4.15]) and 6.36 ranks (CI [4.86, 7.91]). In particular, CORA-aware rewriting reduces MeanRD from +6.33 to -0.03 while increasing R@5 from 37.33% to 40.02%.

However, this improvement is not uniform across individual items or query forms. CORA-aware rewriting improves item-level MeanRD for 377 of the 500 items, while 118 items worsen and 5 remain unchanged. At the query-form level, Key Phrase is the only exception in MeanRD, increasing from +3.03 to +3.78, although its R@5 still improves.

Despite these changes in retrieval performance, the relative ordering of the diagnostic associations remains consistent across all three conditions. Raw query movement remains only weakly associated with RD, whereas TAL and TBMD show stronger associations. For example, under CORA-aware rewriting, $r(\Delta_{\text{move}}, \text{RD}) = 0.020$, compared with $r(\text{TAL}, \text{RD}) = 0.546$ and $r(\text{TBMD}, \text{RD}) = 0.683$. These results show that query rewriting can improve retrieval performance degraded by reformulation at the pipeline level, while the TAL/TBMD and movement diagnostic pattern persists after preprocessing.

N Prompt

N.1 Query Prompt Definitions

Key Phrase Prompt

System Prompt

You are an expert at optimizing audio captions for search engines. Analyze the given caption and compress it into a tight Key Phrase.

Constraints

1. Strictly NO complete sentences (subject + verb). You must use a noun phrase format.
2. Remove unnecessary prepositions and conjunctions.
3. Convert long background descriptions into concise modifiers.
4. Preserve the texture and action of the sound using present participles.
5. Avoid conjunction-heavy structures.

User Prompt

Source Caption: {caption}

Key phrase:

Statement Prompt**System Prompt**

You are an expert at objectively describing audio scenarios. Analyze the given caption and write a full descriptive statement.

Constraints

1. Generate a grammatically complete sentence.
2. Convert stiff noun phrases into natural verbal structures.
3. Preserve simultaneous/background events using conjunctions.
4. End the sentence with a period.

User Prompt

Source Caption: {caption}

Statement:

Question Prompt**System Prompt**

You are an expert at generating questions to verify audio content. Analyze the given caption and create a natural yes/no question.

Constraints

1. Preserve important audible content.
2. Convert the caption into a fluent yes/no question.
3. Avoid overly mechanical wording.
4. End the query with a question mark.

User Prompt

Source Caption: {caption}

Question:

Command Prompt**System Prompt**

You are an expert at crafting precise instructions for search systems or agents. Analyze the given caption and generate a direct command.

Constraints

1. Start directly with an action verb.
2. Use concise search-oriented directives.
3. Preserve important audible content.
4. End the query with a period.

User Prompt

Source Caption: {caption}

Command:

Indirect Prompt**System Prompt**

You are an expert in highly polite and conversational communication. Create an indirect or polite request asking to find the sounds described in the caption.

Constraints

1. Begin with a natural polite expression.
2. Smoothly connect the request to the sound retrieval task.
3. Preserve important audible content.
4. Use appropriate punctuation depending on sentence form.

User Prompt

Source Caption: {caption}

Indirect:

Output Schema

For *question*, *command*, and *indirect*, the model additionally follows:

Output Schema

Return only a valid JSON object with exactly these string fields:

- "answer": the final query text.
- "explanation": a concise explanation of how you chose the final query.

In "explanation", first generate {N} plausible starting words or phrases for the final query, then choose a final starting word or phrase that is *not* in that generated list. The "answer" must start with that non-listed

choice. Explain briefly that the final query starts with the non-listed choice.

N.2 Real-User Query Classification Prompt

Real-User Query Classification Prompt

System Prompt

You are an expert evaluator for classifying user queries by linguistic form.

Your task is to classify the given user question into exactly one of the following six labels:

1. Key Phrase
2. Statement
3. Question
4. Command
5. Indirect
6. None

Classify based only on the linguistic form and user intent of the text. Do not judge factuality.

Query Type Guidelines

1. Key Phrase

A compressed noun-phrase-style query.

Characteristics:

- Not a complete sentence.
- No explicit subject-verb sentence structure.
- Uses concise phrase composition.
- Often omits function words.

Example: “Rain pattering on metal with distant thunder”

2. Statement

A declarative sentence.

Characteristics:

- Complete sentence with subject and verb.
- States or describes something directly.
- Usually ends with a period.

Example: “There is the sound of rain pattering on metal while thunder rumbles in the distance.”

3. Question

A direct question asking for information or whether something is true.

Characteristics:

- Uses interrogative form or question wording.
- Often starts with words such as “What”, “Why”, “How”, “Can”, “Does”, “Is”, “Are”, or “Who”.
- Usually ends with a question mark.

Example: “Is there a recording of rain pattering on metal while thunder rumbles in the distance?”

4. Command

A direct instruction or imperative request.

Characteristics:

- Starts with an imperative verb such as “Find”, “Search”, “Locate”, “Retrieve”, “Tell”, “Explain”, or “List”.
- No explicit subject is needed.
- Directly tells a system or person to do something.

Example: “Search for a recording of rain pattering on metal while thunder rumbles in the distance.”

5. Indirect

A polite, softened, or indirect request.

Characteristics:

- Uses expressions such as “Could you please”, “I would appreciate it if you could”, “I was wondering if you might”, or “Would it be possible”.
- Requests an action without using a direct imperative command.

Example: “Could you please locate an audio file featuring rain pattering on metal while thunder rumbles in the distance?”

6. None

Use this label only when the text does not fit any of the five labels above.

Examples:

- Empty or nonsensical text.
- A fragment that is not a meaningful key phrase.
- Metadata, markup, or text that is not a user query.

User Prompt

User Question: {query}

Query Type:

N.3 Query Preprocessing Prompts

General Rewriting Prompt

System Prompt

You are an expert at rewriting user queries for the audio retrieval process.

User Prompt

Rewrite the following audio search query to improve information retrieval, preserving the original intent and all audible content while improving clarity and adding only relevant acoustic context. Return only the rewritten query.

Input Query: {query}

Output:

CORA-Aware Rewriting Prompt

System Prompt

Instruction: You are an expert at rewriting user queries for the audio retrieval process.

Requirement: Given an audio search query and its query type, your task is to provide a rewritten caption-style query. Please refer to the query rewriting guidelines and examples.

Query Types and Rewriting Guidelines

Question: Convert a yes/no or interrogative query into a direct declarative description of the audible content. Remove expressions such as “Can you hear,” “Does this audio contain,” and “Is there a recording of.”

Command: Convert a direct retrieval instruction into a description of the audible content. Remove imperative and retrieval-oriented expressions such as “Find,” “Search for,” “Locate,” and “Retrieve.”

Indirect: Convert a polite or softened retrieval request into a direct description of the audible content. Remove politeness markers, conversational wrappers, hedging, and indirect request expressions.

Key Phrase: Preserve the compressed phrase structure while rewriting it as a concise and natural audio caption.

Statement: Preserve the directly described sound scene while removing redundant wording and making the description concise.

Examples

Input Query: Is there a recording of rain pattering on metal while thunder rumbles in the distance?

Action: Question

Output: [“Rain pattering on metal while thunder rumbles in the distance.”]

Input Query: Search for a recording of glass breaking followed by people shouting.

Action: Command

Output: [“Glass breaking followed by people shouting.”]

Input Query: Could you please locate an audio file featuring waves crashing against rocks?

Action: Indirect

Output: [“Waves crashing against rocks.”]

Input Query: Birds chirping, light wind, distant traffic

Action: Key Phrase

Output: [“Birds chirping with light wind and distant traffic.”]

Input Query: A train passes while a warning bell rings.

Action: Statement

Output: [“A train passing while a warning bell rings.”]

User Prompt

Input Query: {query}

Action: {query_type}

Output: